Practice workbook · Define diffusion, osmosis and active transport

Definitions of the three transport processes, as written answers, matching lines, cloze passages or describing active transport.

9 question parts: 1 worked-example retry and 8 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-DEFINE-MEMBRANE-TRANSPORT

1. Question 2 · part b · 3 marks

November 2021 Paper 1

Worked-example retry · Attempt part b only.

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2 Molecules move in and out of cells by three methods.

These three methods are:

- diffusion
- active transport
- osmosis

(a) Give a definition of diffusion.

(2)

(b) Give a definition of active transport.

(3)

(c) Give a definition of osmosis.

(2)

(Total for Question 2 = 7 marks)

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1. Question 2 · part b · 3 marks · Working and answer

Working and answer

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2. Question 5 · part a · 2 marks

June 2019 Paper 2R

Further practice · Attempt part a only.

5 Substances move in and out of cells by diffusion, osmosis and active transport.

(a) Describe the process of active transport.

(2)

(b) Water moves in and out of potato cells by osmosis.

Describe a method to determine the concentration of the contents of potato cells.

A large potato is available and also sucrose solutions of various concentrations.

(5)

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2. Question 5 · part a · 2 marks · Working and answer

Working and answer

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3. Question 8 · part a.i · 2 marks

November 2020 Paper 1R

Further practice · Attempt part a.i only.

8 (a) CFTR is a protein that is found in cell membranes.

This protein is responsible for transporting sodium ions across cell membranes using active transport.

(i) Describe the process of active transport.

(2)

(ii) In some people a DNA mutation causes the production of a faulty CFTR protein.

This results in cystic fibrosis.

Explain how an individual can inherit cystic fibrosis from their parents.

(2)

(iii) The faulty CFTR protein is unable to bind to the endoplasmic reticulum in the cell following protein synthesis.

Suggest how this might affect the CFTR protein.

(2)

(iv) The CFTR protein is made up of 1408 amino acids.

Calculate the number of nucleotides found in one strand of the CFTR gene.

(2)

number of nucleotides = ...

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3. Question 8 · part a.i · 2 marks · Working and answer

Working and answer

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4. Question 2 · part a · 2 marks

November 2021 Paper 1

Further practice · Attempt part a only.

2 Molecules move in and out of cells by three methods.

These three methods are:

- diffusion
- active transport
- osmosis

(a) Give a definition of diffusion.

(2)

(b) Give a definition of active transport.

(3)

(c) Give a definition of osmosis.

(2)

(Total for Question 2 = 7 marks)

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4. Question 2 · part a · 2 marks · Working and answer

Working and answer

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5. Question 2 · part c · 2 marks

November 2021 Paper 1

Further practice · Attempt part c only.

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2 Molecules move in and out of cells by three methods.

These three methods are:

- diffusion
- active transport
- osmosis

(a) Give a definition of diffusion.

(2)

(b) Give a definition of active transport.

(3)

(c) Give a definition of osmosis.

(2)

(Total for Question 2 = 7 marks)

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5. Question 2 · part c · 2 marks · Working and answer

Working and answer

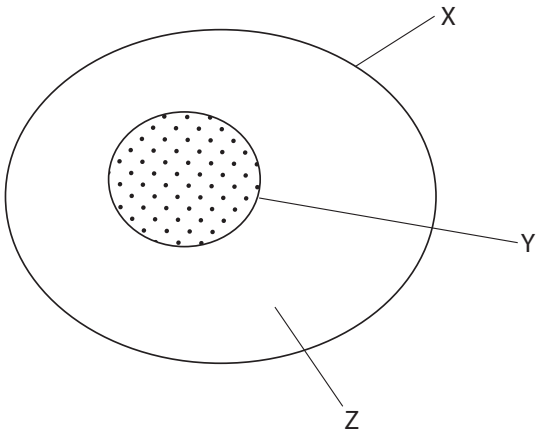
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6. Question 1 · part b · 3 marks

June 2022 Paper 1R

Further practice · Attempt part b only.

1 (a) The diagram shows a human body cell.



(i) Name the parts X, Y and Z.

(3)

X .....

Y .....

Z .....

(ii) Name one structure in the cell that is not shown in the diagram.

(1)

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(iii) Explain how body cells are organised to form organs.

(2)

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6. Question 1 · part b · 3 marks

June 2022 Paper 1R

Further practice · Attempt part b only.

(b) The boxes list three processes that transport substances in and out of cells.

Complete the diagram by drawing one straight line from each process to its correct definition.

(3)

Process	Definition
	requires energy to move molecules against a concentration gradient
osmosis	
	water moves from a high water potential to a low water potential
diffusion	
	water moves from a low water potential to a high water potential
active transport	
	molecules move down a concentration gradient

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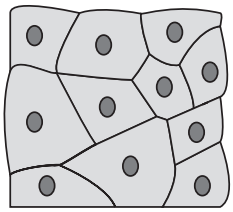
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6. Question 1 · part b · 3 marks

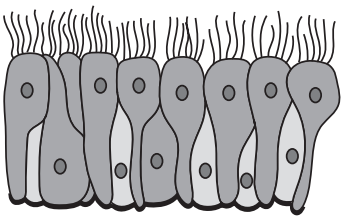
June 2022 Paper 1R

Further practice · Attempt part b only.

(c) (i) The diagram shows two types of epithelial tissue, A and B.



Tissue A



Tissue B

(Source: © Timonina/Shutterstock)

Give the name of each type of epithelial tissue.

(2)

A .....

B .....

(ii) An epithelial cell has an actual width of 60 micrometres (μm).

The cell is magnified 40 times.

Calculate the image width of this cell in mm.

[1 mm = 1000 μm]

(3)

image width = ... mm

(Total for Question 1 = 14 marks)

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6. Question 1 · part b · 3 marks · Working and answer

Working and answer

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7. Question 3 · part a.iii · 1 mark

January 2023 Paper 2

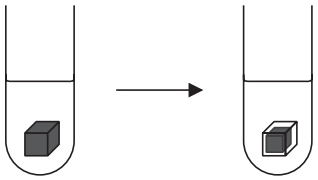
Further practice · Attempt part a.iii only.

3 A student investigates diffusion using agar cubes and an acid.

The agar cube contains an alkali and a pH indicator.

The indicator is pink in alkali and colourless in acid.

Acid diffuses into the cube, causing the cube to change from pink to colourless from the edge.



agar cube when first placed in acid

agar cube starts to become colourless

(a) The cube the student placed in the acid had a side length of 2 mm.

(i) Calculate the surface area to volume ratio of the cube.

Your answer should be in the form n : 1

(3)

surface area to volume ratio = ...

(ii) The cube took 1 minute and 32 seconds to become colourless.

Which of these shows the time in seconds?

(1)

- ☒ **A** 62
- ☒ **B** 82
- ☒ **C** 92
- ☒ **D** 102

(iii) State what is meant by the term **diffusion**.

(1)

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7. Question 3 · part a.iii · 1 mark

January 2023 Paper 2

Further practice · Attempt part a.iii only.

(iv) Explain the effect of increasing the surface area to volume ratio on the time taken for the cube to become colourless.

(2)

(b) A student wants to investigate the effect of temperature on the rate of diffusion into the agar cube.

(i) Describe a method for this investigation using the apparatus shown in the diagram.

(2)

(ii) Give two variables the student would need to control in their investigation.

(2)

1 .....

2 .....

(Total for Question 3 = 11 marks)

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7. Question 3 · part a.iii · 1 mark · Working and answer

Working and answer

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8. Question 1 · part whole · 6 marks

June 2023 Paper 1

Further practice · Attempt part whole only.

1 Use information from the box to complete the sentences about the movement of molecules.

(6)

ADP	ATP	against	along	down	fully
gases	selectively	volume	water		

Molecules can move in and out of cells by osmosis, diffusion and active transport.

Osmosis is the movement of ..... across a ..... permeable membrane.

Diffusion allows the passage of molecules ..... a concentration gradient. The speed of movement is affected by the surface area to ..... ratio of the cell.

Active transport requires the use of energy stored in the form of ..... and allows molecules to move ..... a concentration gradient.

(Total for Question 1 = 6 marks)

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8. Question 1 · part whole · 6 marks · Working and answer

Working and answer

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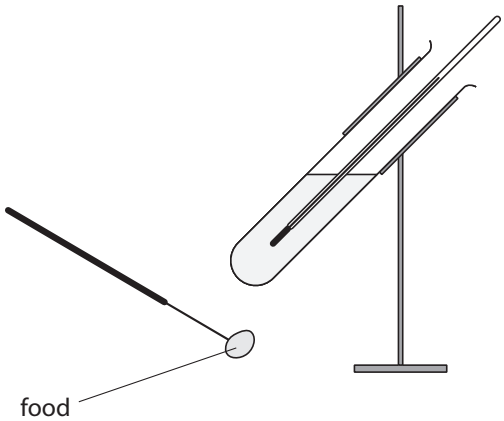
9. Question 7 · part b · 2 marks

November 2024 Paper 1

Further practice · Attempt part b only.

7 In a healthy person, the amount of energy in the daily diet should equal the amount of energy used by the body.

(a) The equipment shown can be used to measure the amount of energy in food.



Describe how this equipment could be used to compare the energy content of different foods.

(6)

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9. Question 7 · part b · 2 marks

November 2024 Paper 1

Further practice · Attempt part b only.

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(b) Describe a named method of transport that uses energy to move molecules across a cell membrane.

(2)

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(Total for Question 7 = 8 marks)

Working and answer

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